MOHAMED SHERIFF MOHAMED WASEEM

BSc. In Electrical & Electronics Engineering

PROFESSIONAL PROFILE

Electrical and Electronics Engineering graduate from SLIIT, with academic coursework completed and awaiting convocation in 2026. Hands-on industrial training experience in water treatment plant operations, instrumentation, motor and pump systems, relay operations, IoT automation, embedded systems, PCB design, and dashboard-based monitoring. Strong interest in electronics, PLC/SCADA, pneumatics, electro-pneumatics, industrial automation, and IoT-based control systems, with practical exposure to ESP32, Raspberry Pi, MQTT, sensors, actuators, relays, and control panels.

EDUCATION

BSc (Hons) in Electrical and Electronic Engineering| Sri Lanka Institute of Information Technology (SLIIT)
Malabe, Sri Lanka | *Graduating 2026* | **Status:** Academic coursework completed; awaiting convocation.

G.C.E. Advanced Level (A/L) - Physical Sciences | Zahira College Kalmunai
Sri Lanka | *Dec 2018*

EXPERIENCE

Assistant Electrical Engineering Intern Zhongnan SZSG JH Joint Venture Company, Ambathale Water Treatment Plant Project
Ambathale, Sri Lanka | *Jun 2024 - Sep 2024*

- Gained practical exposure to pump house operations, flow/pressure monitoring, remote monitoring, and inspection log keeping.
- Observed maintenance activities including insulation testing, pump vibration, cooling water, and surge vessel level maintenance.
- Developed knowledge of instrumentation, motor troubleshooting, relays, control panels, VFD fundamentals, and HSE practices.

Research & Development Engineering Intern Divor Automations (Pvt) Ltd
Pannipitiya, Sri Lanka | *Jun 2025 - Sep 2025*

- Worked on IoT device development, PCB design, embedded programming, panel board assembly, and industrial automation prototypes.
- Supported Hydroponic Grow Box activities including sensor calibration, automatic lighting/irrigation control, and ESP32-based IoT integration.
- Designed and tested PCB layouts using EasyEDA while gaining practical experience in soldering, continuity testing, hardware debugging, MQTT communication, and IoT dashboard concepts.

PROJECTS

Real-Time Automated Hydroponic Cultivation System using IoT and Sensors | Final Year Project | *2026*

- Implemented a multi-node IoT automation system using ESP32, Raspberry Pi 5, Mosquitto MQTT, backend server, and web/mobile dashboard for real-time monitoring and control of sensors, actuators, dosing, pump/valve operation, and safety functions.

Armyworm Pest Detection and Ultrasonic Repeller System | Design Project | *2025*

- Designed an intelligent agricultural pest detection and repelling prototype using Raspberry Pi 5, camera-based image detection, Python/OpenCV, TensorFlow Lite, servo scanning, GPIO control, and an ultrasonic Repeller circuit for non-chemical pest control.

TECHNICAL SKILLS

Programming & Embedded:	Embedded C / Arduino, Python, Java
Microcontrollers & Platforms:	ESP32, Arduino boards, Raspberry Pi 5, STM32, NodeMCU, ESP8266
Automation & Control:	SCADA/HMI exposure, PLC fundamentals, relay logic, VFD fundamentals, control panels, motor/pump systems
IoT & Networking:	MQTT, Mosquitto broker, Wi-Fi, dashboards, remote monitoring, Cloudflare Tunnel, VNC, Docker basics
PCB & Electronics:	EasyEDA, schematic design, PCB layout, Gerber generation, SMD/THT soldering, hot plate soldering, continuity
Testing Equipment:	Digital multimeter, oscilloscope, power supply, insulation resistance tester, spectrum analyzer exposure
Sensors & Actuators:	pH, TDS, ultrasonic level, DHT sensors, pressure/flow/level instrumentation, solenoid valves, relays, pumps

REFERENCES

Available upon request.

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